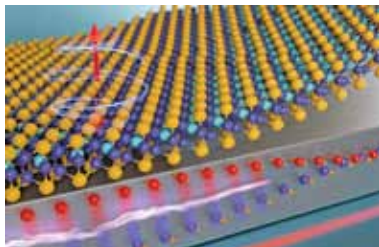


(Image: Courtesy of the researchers)

the spool holder. The result is more compact devices with stronger magnetic fields than other 3D-printed solenoids. Although not as powerful as traditional solenoids, these could be useful in small sensors or soft robots. The team aims to enhance performance by exploring better materials and refining temperature control to minimise defects.

Low-energy magnetic computing for efficient processors and memories

MIT researchers have developed experimental computer memories and processors using two-dimensional magnetic materials, just a few atoms thick, that consume much less energy than traditional silicon devices. The team dem-



This illustration shows electric current being pumped into platinum (the bottom slab), which results in the creation of an electron spin current that switches the magnetic state of the 2D ferromagnet on top. The coloured spheres represent the atoms in the 2D material (Image: Courtesy of the researchers)

onstrated precise control of a van der Waals magnet at room temperature, a significant achievement since these thin magnetic materials usually require low temperatures for control. They achieved room-temperature magnetic switching using electrical current pulses, crucial

for binary code representation and nonvolatile computer memories. The material used, iron gallium telluride, retains its magnetism at higher temperatures and requires lower electrical current for switching, making it more energy-efficient. Advancement for faster computers, nonvolatile memories, and more efficient processors

Large-scale production of stable, high-performance solar cells

Researchers at São Paulo State University (UNESP) in Bauru, Brazil, have advanced the efficiency and stability of perovskite solar cells by incorporating MXenes, a class of two-dimensional materials similar to graphene. This innovation enhances the power conversion efficiency of the cells from 19% to 22% and triples their stability. MXenes,

composed of transition metals, carbon, nitrogen, and functional groups like fluoride, oxygen, or hydroxyl, offer high electrical conductivity and thermal stability. Initially aimed at mitigating performance decline due to an insulating passivation layer, the study now opens avenues for large-scale industrial production of stable, high-performance solar cells. It outlines a viable method for mass production and various characterisation techniques, marking a significant step towards sustainability goals and positioning Brazil as a leading solar cell manufacturer.

AI-driven efficiency in robotic warehouses

MIT researchers have developed an AI-enhanced method to improve efficiency in large robotic warehouses, where hundreds of robots transport items for packing and shipping.



A group of MIT researchers who use AI to mitigate traffic congestion applied ideas from that domain to tackle the problem of multiple robots in a warehouse setting (Credit: iStock)

Using a deep-learning model, they predict the best areas to decongest and divide robots into smaller groups for faster resolution, making their method nearly four times quicker than traditional approaches. The AI model focuses on areas with the highest potential to reduce total travel time

by analysing smaller groups of robots and predicting which group's coordination would most improve overall efficiency. The neural network manages robot groups by understanding their complex interactions, such as potential path crossings, and simplifies computations by encoding constraints only once. In simulations, this learning-based approach proved up to four times faster than non-learning methods, even accounting for the neural network's extra computational effort.

New wearable sensor enables communication through hand and finger movements

Researchers from Beijing Normal University, Sun Yat-sen University, and Guilin University of Electronic Technology have developed a new wearable sensor that enables communication through hand and finger movements. The sensor combines a flexible silicone elastomer, polydimethylsiloxane (PDMS), with an optical component called a fibre Bragg grating (FBG) for accurate movement detection. Designed for comfort and long-term wear, the sensor can be applied to various body parts and is adaptable to individual users through precise calibration.